

PETER HUMULOCK

Senior Software Engineer · Distributed Systems & Platform Engineering
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Active Top Secret Clearance

SUMMARY

Software engineer with 10+ years building scalable, data-driven systems end to end — distributed backends, event-driven architecture, and cloud infrastructure. Depth in Go and JVM services, messaging and streaming, and PostgreSQL performance, with a record of setting security and data-governance standards in cleared environments.

EXPERIENCE

Technical Marketing Engineer | Synadia · Baltimore, MD (Remote) April 2025 – August 2026

- Designed and built reference architectures and production-grade demo applications on NATS.io and Synadia Cloud, demonstrating event-driven patterns including pub/sub, JetStream persistence, and key-value and object stores.
- Built AI-assisted developer tooling and agent integrations — MCP servers and custom skills across Claude and OpenAI Codex — applying working knowledge of LLM architecture and context management to real workflows.
- Produced the developer documentation, technical guides, and video demos teams used to adopt NATS in production.
- Worked across the stack in Go, TypeScript, and hypermedia frameworks to demonstrate low-latency application patterns from broker to browser.

Senior Research Engineer | Two Six Technologies · Baltimore, MD (Remote) September 2020 – April 2025

- Architected scalable backend services with Vert.x and HyperExpress on a Hazelcast-backed cluster, bringing up live data streams with CDC to propagate database changes in real time across services for resilient, distributed state.
- Implemented secure authentication workflows using OAuth2 and PKCE, and designed and enforced Role-Based Access Control for granular, auditable access to protected resources.
- Established data governance practices in compliance with ISM standards, defining the controls the team built against.
- Optimized PostgreSQL schemas and query performance; integrated WebSockets into Ember.js for real-time updates.
- Mentored junior engineers and raised the team baseline on security and API design — consistent HTTP semantics, versioning, and error handling.

Software Development Engineer | U.S. Engineering Solutions · Baltimore, MD March 2018 – September 2020

- Built real-time weather monitoring dashboards in React and Redux, reducing data load times by 30%.
- Developed Spring Boot and Hibernate services backed by Redis to cache and serve high-frequency weather data.
- Designed an AWS ingestion pipeline using Lambda for real-time processing and S3 for durable, scalable storage.

Earlier Eutality Group (2017–2018) · Mannakee Group (2016–2017) · e-Management (2014–2016) — C#/ .NET physics simulation, AngularJS and SharePoint applications for high-security environments, and a Java web crawler with D3.

SELECTED PROJECTS

Vert.x Hazelcast Sensor Cluster · Java 21 · Vert.x · Hazelcast · Datastar · Docker

A single clustered event bus spanning three JVMs, with a Hazelcast AsyncMap as the only data store behind cluster-wide locks. Sensors deploy and tear down at runtime; readings reach the browser over SSE.

Datastar NATS Tic-Tac-Toe · Go · NATS JetStream KV · SSE · CQRS

The server owns all state: moves are validated into JetStream KV and rendered markup is pushed to both players over SSE — no client-side application logic, no JSON API. CQRS separates writes from KV-watch reads; presence heartbeats expire orphaned games.

xsym · Rust · tree-sitter · SQLite · MCP

A cross-language structural code index served over MCP: normalization collapses `ConsumerConfig`, `consumer_config` and `jsConsumerConfig` to one key, so an agent can ask where a concept lives across Go, Rust and Python. Adding a language is a tree-sitter query file, not a visitor; 52,251 symbols across 40 repos, no-op re-index in 0.7s.

TECHNICAL SKILLS

Languages Go, Java, JavaScript, TypeScript, C#, SQL

Messaging NATS.io / JetStream, RabbitMQ, Redis Streams, Hazelcast, WebSockets, gRPC

Architecture Event-driven systems, microservices, event sourcing, CQRS, SAGA, serverless

Backend Vert.x, Spring Boot, Node.js (Express, Fastify), ASP.NET Core, REST, GraphQL

Data PostgreSQL, Redis, MongoDB, MySQL, schema design, query and index optimization

AI & Agents LLM integration, MCP servers, agent skills and tooling, prompt and context design

Frontend & Mobile React, Vue.js, Svelte, Angular, Ember.js, Datastar, TailwindCSS, Swift / iOS, Android

Cloud & DevOps AWS (Lambda, S3), Docker, Kubernetes / K3s, Fly.io, GitHub Actions, GitLab CI

Security OAuth2 / PKCE, JWT, Role-Based Access Control, ISM compliance

Testing Playwright, k6, Jest, JUnit

EDUCATION

B.S., Computer Science | University of Maryland, College Park, MD 2010 – 2014

A.S., Computer Science | College of Southern Maryland, La Plata, MD 2008 – 2010